COLLEGE OF SCIENCE
CHEMISTRY
VIRGINIA TECH

General Chemistry 1036
Spring 2026

Time and Location of Lectures: MWF 12:20 PM–1:10 PM in Davidson 281

Instructor:	Office:	Email:
Dr. Shamindri Arachchige	Davidson 109	arachsm@vt.edu

Learning Assistants:	Email:
Brooke Laine	brookelaine@vt.edu
Kiera Bowden	kierab23@vt.edu

Student Hours
Mondays from 1:30 PM – 3:00 PM in the Davidson 220 classroom by Dr. Arachchige
Tuesdays from 4:00 – 5:00 PM in the HAHN South Atrium Study Area by Brooke
Wednesdays from 1:30 PM – 3:00 PM in the Davidson 220 classroom by Dr. Arachchige
Thursdays from 11:00 AM – 12:00 PM in the HAHN South Atrium Study Area by Kiera
Fridays from 1:30 PM – 3:00 PM in the Davidson 220 classroom by Dr. Arachchige

Other hours by appointment. Email arachsm@vt.edu

CHEM 1036 Problem Solving Studio: You are enrolled in a Chem 1036 section that includes problem-solving studio sessions. In addition to attending lectures, you are required to participate in mandatory studio sessions held on Thursday afternoons. The Course Reference Number (CRN) you selected assigns you to a specific studio section. Attendance at your assigned studio each week is required to receive credit, and you may only attend the studio in which you are enrolled. For additional details about studio sessions, please refer to pages 4–5 of the course syllabus.

Problem Solving Studio Location and Times (**all are held on Thursday afternoons**):

CRN	TIME (PM)	LOCATION	INSTRUCTOR	EMAIL
12850	6:00 – 6:50	PAM 1001	Ava Hamilton	avah09@vt.edu
12851	6:00 – 6:50	PAM 2001	Sneha Sapkota	snehas22@vt.edu
12853	5:00 – 5:50	DAV 125	Closed	
12856	2:00 – 2:50	DAV 220	Tamas Gal	tamasgal@vt.edu
12858	5:00 – 5:50	DAV 201	Closed	
12866	3:30 – 4:20	DAV 201	Ava Hamilton	avah09@vt.edu
12876	3:30 – 4:20	DER 3092	Melina Nejadian	melinan@vt.edu
12882	5:00 – 5:50	PAM 2002	Closed	
12883	6:00 – 6:50	ROB 116	Maryam Imran	maryamimran@vt.edu
12886	3:30 – 4:20	PAM 2001	Closed	
12887	6:00 – 6:50	DER 3092	Ola Khashan	olakhashan@vt.edu
12889	2:00 – 2:50	DAV 320	Carson Hotchkiss	achotc80@vt.edu
12890	6:00 – 6:50	PAM 2028	Liya Eshetu	liyae@vt.edu

Students that have not successfully completed CHEM 1035 (or CHEM 1055) may not take CHEM 1036 or CHEM 1046 (lab) and will be dropped from the class. Students cannot take CHEM 1035 and CHEM 1036 in the same semester.

Text: Chemistry: The Molecular Nature of Matter and Change 10th Ed., Martin S. Silberberg and Patricia G. Amateis. Chapters 11-13 and 16 – 21 will be covered. Please see pages 16-21 for information on purchasing the textbook.

Required Calculator: The ONLY calculators that may be used on tests and quizzes are TI-30X IIS or TI-30XS Multiview by Texas Instruments. Calculators will not be provided, so you must bring your own to all quizzes, tests, and the final exam. Use of any other calculator constitutes a violation of the Honor Code.

iClicker: You are required to have an iClicker 2 student remote. This remote can be purchased from the VT Bookstore or you can rent or purchase it from the following website:
Macmillan Student Store Please see pages 13-17 for information on purchasing a remote.

If you cannot afford to purchase the textbook or other learning materials, please go to Student Emergency Fund Contact Form

Tests: Four tests and a cumulative final exam will be given throughout the semester as scheduled below. The testing classroom locations will be announced. Please see pages 9-13 for more information about the Chem 1036 testing policies.

TEST	DATE	CHAPTERS	TIME (ET)
Test 1	February 19th (Thursday)	11, 12, 13	8:00-9:30 PM
Test 2	March 19th (Thursday)	16, 17	8:00-9:30 PM
Test 3	April 16th (Thursday)	18, 19	8:00-9:30 PM
Test 4	May 1st (Friday)	20, 21	6:00-7:30 PM
Final Exam (Cumulative)	May 9th (Saturday)	11, 12, 13, 16,17,18, 19, 20, 21	10:05 AM-11:50 AM

Test Absences: You are expected to take all tests on the scheduled dates unless your absence is excused due to an extended illness, emergency, or university-approved conflict. Only university-verified absences, including documented schedule conflicts, severe illness, family emergencies, or Dean of Students approved absences, will be accepted as valid excuses for missing a test or exam.

All requests for excused absences must be emailed to me as soon as possible and must include your name, class time, and the reason for your absence. For absences due to a university schedule conflict, requests must be submitted at least three days prior to the test date and must include documentation from a coach or academic advisor. You are responsible for reviewing your schedule now and notifying me of any conflicts.

Only students with approved, university-verified absences are eligible to take a makeup test. Students who do not qualify for a makeup test will receive a score of zero for the missed exam. Makeup tests will be administered from 8:00–9:30 PM on 2/23, 3/23, 4/20, and 5/04/2026 (locations to be announced).

There is no makeup or alternate date for the final exam, and remote testing is not permitted for any test or the final exam. Makeup tests will not be returned or published; therefore, students who take a makeup test will not have access to their exam responses and should use the regularly scheduled class exams to assess their progress.

Note: Students are strongly advised not to make travel plans until the conclusion of final exam week (May 14, 2026) in case exams must be rescheduled due to weather or other unforeseen circumstances.

Dean of Students: If you have an extended or serious illness, a death in your family, or other significant personal difficulty that will prevent you from attending class, meeting your assignment deadlines, or cause you to miss quizzes or exams, please notify the Dean of Students. The Dean of Students will secure your personal verification and private documentation such as notes from doctors, funeral directors, etc. The Dean of Students will also notify all of your course instructors of your situation with a date range for your verified absence. This endorsement from the Dean of Students helps an instructor extend accommodations to you with less concern for unfairness toward your classmates. A verified absence does not guarantee you an extension or makeup assignment. If you experience an absence longer than two weeks, please contact your instructor. There are practical limits on academic accommodations. Some situations are severe enough to warrant other forms of academic relief such as course withdrawal. You can contact the dean of students on 540-231-3787.

Classroom Attendance: Class attendance is mandatory. But, if you must self-isolate, quarantine, or you are exhibiting any symptoms of illness, you must not attend class. There are no makeup assignments for iClicker responses. You are expected to use your dropped iClicker scores responsibly to account for only important and unavoidable absences.

Grading System:

Four Tests: 56% total (14% each test)
Cumulative final exam: 14%
ALEKS Online Assignments (two types of assignments: Modules and Homework): 14%
iClicker Average: 10%
Problem Solving Studio Quizzes: 6%

Overall Average = (Test average × 0.56) + (Cumulative final exam × 0.14) +
(ALEKS average × 0.14) + (iClicker average × 0.10) + (Studio quiz average × 0.06)

Note: When computing your grade
- <u>none</u> of the test scores are dropped.
- the final exam score is <u>not</u> dropped.
- the lowest five ALEKS scores are dropped. (See pages 17-21 for instructions on how to register for online homework).
- The lowest six iClicker performance days (for the MWF class meetings) or the lowest four iClicker performance days (for the MW class meetings) are dropped. (See pages 13-16 for instructions on how to register your iClicker and other related information).
- The lowest three studio quiz scores are dropped. (See pages 4-5 for more information about problem solving studios and studio attendance).

Grading Scale:

≥ 93%	≥ 90%	≥ 87%	≥ 83%	≥ 80%	≥ 77%	≥ 73%	≥ 70%	≥ 67%	≥ 63%	≥ 60%	≥ 0%
A	A–	B+	B	B–	C+	C	C–	D+	D	D–	F

Note: Grade totals in Canvas will not be visible until the course has been completed. When some grades are still missing or pending, the Grade Total displayed is not accurate and is not a correct reflection of your current standing.

Students who wish to estimate the final exam score needed to achieve a desired letter grade should use the Overall Average formula on page 3 to calculate approximately what final exam score is required to reach your desired course average.

Web Page: http://canvas.vt.edu (Use Chrome with Canvas). Here you will find:

Announcements tab: All important course updates will be posted via Canvas Announcements. You are required to read all announcements.

To ensure you receive them promptly, check your Canvas notification settings:
1. Go to Canvas → Account → Notifications
2. Make sure “Announcements” is set to “Notify me right away”.

Files tab:

- Syllabus and Online Homework Information folder: The class syllabus and online homework instructions are here.
- Lecture Outlines folder: Lecture outlines are available here. Use them to take notes, either on the printed copies or by making annotations on your tablet computer.
- Test Solutions folder-Test solutions will be here a few days after each test.
- Worksheets folder- Worksheets for studio are here. Print and bring them to your problem-solving studios or use your tablet for annotations. Solutions are available after each studio.
- Quiz Solutions folder- The solutions to quizzes are here a few days after each quiz.

Quizzes tab: Click on the Quizzes tab to access the tests and the final exam. Also click on the quizzes tab to access the online practice tests given in previous semesters.

Grades tab: Test and final exam scores, quiz scores, online homework scores, and the overall iClicker average will be populated here.

Problem Solving Studio Information: You are enrolled in a Chem 1036 section that involves a problem-solving studio. In addition to the lectures, you will also have mandatory studio sessions on Thursday afternoons. Your Course Reference Number (CRN) places you in a studio section each having about 30 students. The studio will be led by teaching assistant, an upperclassman, that has previously had this course.

Your teaching assistant will structure the time of studio so that you are working with other students on a worksheet that covers principles and concepts discussed during the prior week in the Chem 1036 lectures. You should come to studio class with the worksheet completed to the best of your ability. During studio, your teaching assistant will help clarify any remaining questions you may have on the worksheet. After a time for solving problems on the worksheet, a short quiz will be given on the material covered on the worksheet. In order to be eligible to take the quiz, you must be present from the beginning of the studio meeting. The worksheets will not be graded unless specified otherwise. Solutions to the worksheet and quiz will be posted on Canvas after the studio session. Your quizzes are graded by your teaching assistant. Quiz grades will be posted on Canvas.

Your lowest three quiz grades of the semester will be dropped. If you miss any studios, the zeros received for your absences will be included in the lowest three quiz grades that get dropped. There are no make-up quizzes for the first three missed sessions, regardless of the reason, including excused absences verified by the Dean of Students. To receive full credit for quiz problems, you must show your work.

NOTE: The ONLY calculators that may be used on quizzes and tests are the TI-30X IIS and TI-30XS Multiview by Texas Instruments.

Note on Studio Attendance: Studio attendance is mandatory. However, if you exhibit any symptoms of illness, you must self-isolate or quarantine. The lowest three studio quiz scores will be dropped. If you miss a studio quiz, it will be recorded as a zero, but these zeros will be among the three lowest scores dropped.

You are expected to reserve your dropped scores for important and unavoidable absences. This policy is not intended to allow students to skip studios for leisure. Each studio session provides a valuable learning opportunity that should not be missed. There are no makeup studio quizzes for the first three absences, including those verified by the university. If you anticipate a fourth university-verified absence, you should contact your instructor to request a makeup quiz.

Note: You will only be eligible to take a makeup quiz if all the following requirements are met:

(1) the absence must be your fourth university verified absence
(2) your three previous absences must also be university-verified absences. You must produce the three previous absence verifications at the time of the request for the fourth makeup quiz.
(3) your request is submitted within 24 hours for severe illnesses and three days in advance for university conflicts.

Getting Help with Chem 1036

Plenty of help is available for explaining concepts and working problems:

- Student hours are held by instructor and learning assistant.
- The Chemistry Department provides free tutorial assistance in the Chemistry Learning Center (CLC) located in Davidson 219. No appointment necessary; just walk in. Hours of operation TBA (will open second week of class).
- Alpha Chi Sigma, a chemistry fraternity, provides free tutoring at Davidson 219 in the evenings. I will let you know the hours once those are set.
- The Student Success Center provides free tutoring. Contact them through their web page: www.studentsuccess.vt.edu
- A private tutor list will be posted on the Canvas page soon.
- Free math tutoring by the Math Tutoring Lab
- ALL CHEM 1036 students are invited to any of the help sessions below.

Schedule of Help Sessions for Tests

Prof. Wagner and I will be holding review sessions on the Wednesday before each Chem 1036 Test. You may attend any that permits your schedule.

Test 1 Help Sessions

Day	Time (PM)	Zoom Link or Location	Professor
Wednesday, Feb. 18	5:30-7:00	Davidson 281	Prof. Wagner
Wednesday, Feb. 18	7:30-9:00	https://virginiatech.zoom.us/j/84550580912	Prof. Arachchige

Test 2 Help Sessions

Day	Time (PM)	Zoom Link or Location	Professor
Wednesday, Mar. 18	5:30-7:00	Davidson 281	Prof. Wagner
Wednesday, Mar. 18	7:30-9:00	https://virginiatech.zoom.us/j/84550580912	Prof. Arachchige

Test 3 Help Sessions

Day	Time (PM)	Zoom Link or Location	Professor
Wednesday, April 15	5:30-7:00	Davidson 281	Prof. Wagner
Wednesday, April 15	7:30-9:00	https://virginiatech.zoom.us/j/84550580912	Prof. Arachchige

Test 4 Help Sessions

Day	Time (PM)	Zoom Link or Location	Professor
Wednesday, April 29	5:30-7:00	Davidson 281	Prof. Wagner
Wednesday, April 29	7:30-9:00	https://virginiatech.zoom.us/j/84550580912	Prof. Arachchige

Honor Code: The Undergraduate Honor Code pledge that each member of the university community agrees to abide by states: "As a Hokie, I will conduct myself with honor and integrity at all times. I will not lie, cheat, or steal, nor will I accept the actions of those who do." Students enrolled in this course are responsible for abiding by the Honor Code. A student who has doubts about how the Honor Code applies to any assignment is responsible for obtaining specific guidance from the course instructor before submitting the assignment for evaluation. Ignorance of the rules does not exclude any member of the University community from the requirements and expectations of the Honor Code. Academic integrity expectations are the same for online classes as they are for in person classes. All university policies and procedures apply in any Virginia Tech academic environment. If Virginia Tech has to transition to online classes, using sites and/or apps such as Chegg, CourseHero, GroupMe, Discord, etc. during tests in this class are considered Honor Code violations. Virginia Tech is able to effectively investigate incidents of use of these types of technology as violations, and we will report them as such.

It is understood that turning in any answer sheet or using an iClicker constitutes an agreement to abide by the code. Students who vote for their classmates at any time are violating the Honor Code.

The ONLY calculators that may be used on tests and the final exam are TI-30X IIS or TI-30XS Multiview by Texas Instruments. Use of any other calculator is a violation of the Honor Code.

Asking me to give you a letter grade that you did not earn is considered a violation of the Honor Code.

You will sign an Honor Code pledge on every test and exam in this class.

For additional information about the Honor Code, please visit:
https://www.honorsystem.vt.edu/

To support your academic journey at Virginia Tech, we highly recommend completing the following self-paced modules:

- Academic Integrity Success module (recommended):
  https://canvas.vt.edu/enroll/CE7YK9
- Understanding the Code (recommended after Aug 30):
  https://canvas.vt.edu/enroll/AC7F7C

Faculty are expected to adhere to the policy pertaining to the reporting and adjudication of suspected violations of the Honor Code. Any suspected violations of the Honor Code are reported promptly to the Office of Undergraduate Academic Integrity. The normal recommended sanction for a violation of the Honor Code is an F* as your final course grade. The F represents failure in the course. The "*" is intended to identify a student who has failed to uphold the values of academic integrity at Virginia Tech. A student who receives a sanction of F* as their final course grade shall have it documented on their transcript with the notation "FAILURE DUE TO ACADEMIC HONOR CODE VIOLATION." You would be required to complete an education program administered by the Honor System in order to have the "*" and notation "FAILURE DUE TO ACADEMIC HONOR CODE VIOLATION" removed from your transcript. The "F" however would be permanently on your transcript.

Classroom Conduct: The classroom should provide a focused learning environment, free from unnecessary conversation and distractions. Technology can be a helpful tool when used to enhance learning, but it must be used responsibly:

- Silence or put away devices to avoid audible or visual distractions.
- Use devices only for class-related purposes; non-instructional or unprofessional use may affect your final grade.

- Be mindful of your attention: once distracted, it takes significant time to refocus, which impacts learning. Acknowledge your accomplishments when you maintain focus for extended periods. Practicing digital self-control benefits everyone.

- Handwriting your notes in class is strongly encouraged. Even when devices like laptops are diligently used for typed notetaking, studies show they invite rote transcribing versus mental processing or thinking leading to poorer academic performance. Chemistry is not a subject that is easily typed on a laptop. It requires the student to transcribe multistep calculations and draw detailed figures. Hand-writing your notes in class encourages more efficient learning.

Frequent movement in and out of the classroom or leaving early is disruptive. Class starts on time, and you are expected to arrive promptly and remain seated for the entire session unless an emergency arises.

Services for Students with Disabilities (SSD): Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to disability, including but not limited to ADHD, chronic or temporary medical conditions, deaf or hard of hearing, learning disability, mental health, or vision impairment, please contact the Services for Students with Disabilities (SSD) office (540-231-3788, ssd@vt.edu, or visit ssd.vt.edu). If you have an SSD accommodation letter, please meet with me privately during office hours as early in the semester as possible to deliver your letter and discuss your accommodations. You must give me reasonable notice to implement your accommodation, which is generally 5 business days and 10 business days for final exams.

Inclusion, Diversity, and Basic Needs: If there are aspects of this course that prevent you from learning or that exclude you, please let me know as soon as possible. Together we will develop strategies to meet both your needs and the requirements of the course.

- I will honor your request to address you by your chosen name or personal pronoun. Please advise me of this early in the semester so that I may make appropriate changes to my records.

- If you are a veteran or active-duty military personnel with special circumstances (e.g., upcoming deployments, drill requirements, disabilities) please know that I want to work with you to accommodate your circumstances.

- Virginia Tech provides a variety of free services to support student success. These services are more limited in the summer months, but I will post a document on Canvas that will provide links to free services of which I am aware. I will also provide a tutor list –private tutors typically charge $15-20 per hour.

- If I suspect that you need additional support, I will express my concerns and the reasons for them and remind you of resources that might be helpful to you. It is not my intention to know the details of what might be bothering you, but simply to let you know I am concerned and that help is available.

- If you are facing challenges securing food, housing, or childcare and believe this may affect your performance in the course, you are urged to contact the Dean of Students for support. The Dean of Students can connect you to resources within the local community.

- Any student who has difficulty affording groceries, accessing sufficient food to eat every day, or who lacks a safe and stable place to live, or who lacks child care, and who believes this may affect your performance in this course, is urged to contact the Dean of Students office for support at 540-231-3787 (dos@vt.edu) or complete an interest form to participate in The Market at Virginia Tech (https://foodaccess.vt.edu/).

- The Dean of Students, through The Market at Virginia Tech, offers food options and other resources. There is also a Student Emergency Fund program. If you are comfortable in doing so, please notify your professor or departmental advisor of your situation.

- I respect and support your decision to honor your cultural and religious holidays. If you have religious or cultural observances that coincide with this class, please notify me through email at least two days before the date that is in conflict. Also, the Dean of Students can provide you with an Absence Verification for a planned religious observation. Otherwise, I will assume that you plan to attend all class meetings.

- I want you to feel able to share your life experiences in discussions and in your written work. I want you to trust that I will keep any information you share private. Please be aware that I do have a mandatory reporting responsibility related to my role as a faculty member. This means I am required to share information regarding sexual misconduct or information about a crime that may have occurred on campus with the university Title IX Coordinator.

Chem 1036 Testing: Tests are given online on your laptop, in the testing classroom (location will be announced in class). You must have a laptop meeting the baseline minimum requirements specified: https://www.compreq.vt.edu/specs.html You cannot use an iPad or Chromebook to take tests or the final exam. Students must have administrative privileges on their devices. They must be able to install new software and make operating system settings adjustments without the need for a third party to log into their devices. Operating systems must run in EST time-zone.

Note: If Virginia Tech has to transition to online classes due to Covid, you must have a web camera since our testing system requires one. You cannot take a CHEM 1036 test without a webcam IF we have to go totally online. If your laptop does not come with a web camera, please order an external webcam. Note that a webcam was a computer requirement for students in all majors.

Each test is found in Canvas. You will use Lockdown Browser for testing. You won't be able to access a test that requires LockDown Browser without LockDown browser installed in your laptop.

Download and install the VT version of the Lockdown Browser in your laptop from this link:

https://download.respondus.com/lockdown/download.php?id=776344933

Once you have downloaded the VT version of Lockdown browser, please test that it works by running the “Check Your Computer Settings Quiz- Requires Respondus LockDown Browser” found in quizzes on Canvas. Access the quiz using your regular browser. Clicking on the quiz will launch LockDown browser.

The Thursday night tests start at 8:00 PM EST. The Friday night test starts at 6:00 PM EST. You should arrive at your test location at least 15 minutes before the start time and be ready to choose “Begin Quiz” at the start time. You will have 90 minutes (1.5 hours) for each of the four tests. Please keep in mind that each test is actually written to take 75 minutes and there is an additional 15 minutes budgeted for simple technical issues that one might encounter.

The Final exam starts 10:05 AM on May 9th. Arrive at your exam location at least 15 minutes before the start time and be ready to choose “Begin Quiz” at 10:05 AM. You will have 105 minutes (1 hour and 45 minutes) for the final exam. Note that the final exam is actually meant to take 90 minutes with an additional 15 minutes budgeted for simple technical issues.

Students with SSD Accommodations: Your tests and final exam will start earlier than the class. You will be contacted individually about your tests and final exam times.

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For the test, bring

- a fully charged laptop (no Chromebooks or iPads)
- a device you use for two-factor authentication, e.g. your phone
- A course approved calculator with NO COVER. Covers will be confiscated and not returned
- one or two writing implements (but NO PENCIL CASES)
- your Hokie passport.

Note:

- Access to a power source is not guaranteed but bring your power cord just in case.
- Scratch paper and a periodic table/equation sheet will be provided. Do not bring your own. Having any other types of paper in your possession is considered academic misconduct and will be reported to the Honor System.
- NO BACKPACKS are allowed in testing rooms.
- Having a cell phone in your lap, on your desk, or the floor during a test is considered academic misconduct and will be reported to the Honor System. You may bring a cell phone to the test for two-factor authentication but then you MUST put the cell phone in a pocket, and it must be turned off. Your test proctor will check that your cell phone is turned off before the test. If you see a student using a cell phone during the test, it is your responsibility to notify the test proctor immediately.
- Smartwatch and/or headphones are not permitted during testing.
- No food and drinks will be allowed.

Some of these policies may not apply to students who test with SSD. Ask if you are confused/concerned. If you have a concern, please speak with me as soon as possible.

Before the Test

1. Use the restroom before you begin testing. You will not receive extra time for the time you used for leaving the screen.

2. Check Canvas announcements for your testing location. You may not test outside of your assigned room. The access code will not work.

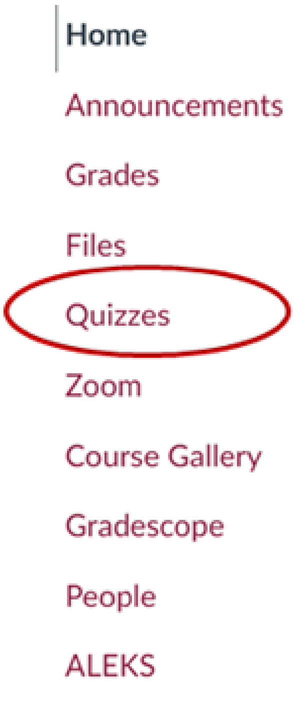
3. Turn your laptop off completely, wait a couple of minutes and then turn it back on.

4. Make sure all other applications are closed.

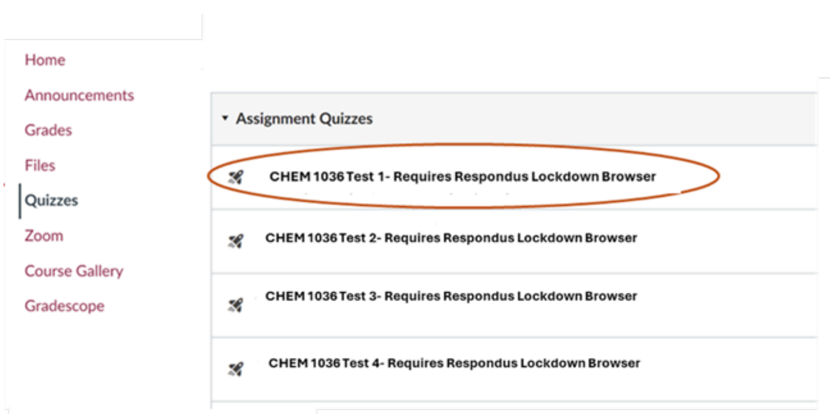
5. Log in to Canvas as normal in your <u>regular browser</u>.

6. Click on your Chem 1036 course.

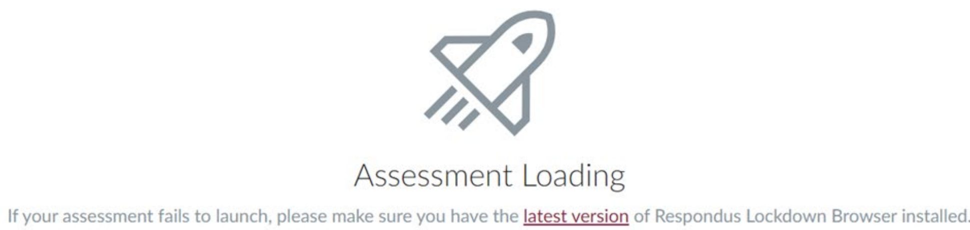
7. Click on Quizzes on the left side of the Canvas site.



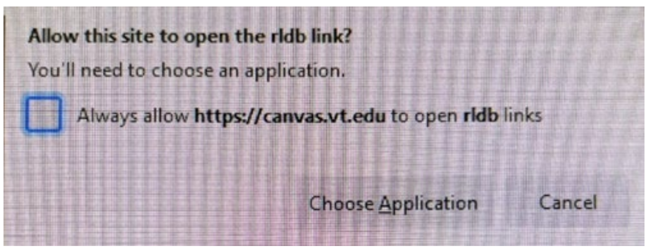
8. Click on the Test you wish to take.



9. A new tab will pop open in your browser that looks like this:



10. A small popup will also open that looks like this



I recommend checking the box so that you do not have to grant access in the future. If applicable, click “Choose Application” and choose Lockdown Browser.

11. Lockdown Browser will open on its own and bring you straight into the Test Start screen. Click on the “Begin” button to begin the Test.

***Note: If you click begin and you get a login error, close Lockdown browser and attempt to open the test again from the start. Restarting your computer resolves most of these issues. You may have to do this multiple times. Don’t panic. Get assistance from your test proctor if you continue to have issues after five attempts of trying.

During the test

1. Write down all of your answers. Not A, B, C, etc., but the actual numerical answers or sentences. For example, write down:

Question
1. SO_4^{2-}
2. 44.9 g
3. $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$ etc.
2. You can go back to questions that you have skipped or to change your answer. Only submit the test once you have answered all of the questions. Be careful that you do not prematurely hit the Submit button! You will not be able to resume a submitted Test attempt! You do NOT get multiple attempts – only one attempt! Once you hit Submit, you are done.

3. After you have completed the test and clicked the “Submit” button, close Lockdown browser.

Note: If you submit your test attempt and are taken back into the test itself, just close Lockdown browser.

4. Once your test is submitted, IMMEDIATELY EXIT the test, log back into the test (start these instructions from the beginning) and view your submitted test attempt (VIEW RESULTS). Check that your answers have been recorded by Canvas by checking them against your written answers.

If you find a major discrepancy between the answers you wrote down and the answers recorded by Canvas, inform the test proctor immediately. The discrepancy must be reported before you leave the testing classroom. Your test proctor will then work to track down any issues with the help of VT's IT Services. Forgetting to enter an answer and/or an incorrect entry are not considered as data loss. We are looking for extremely rare instances of major data loss.

5. Test grades are released usually the day after the test. You will not know your grade immediately upon submission of the test.

After the Test: Leave quietly with respect for your peers who are still testing and exit the building. If you need to wait for some reason, do so outside, or near the exit. Turn in all scratch paper, used or unused, if you leave before 9:00 PM for the Thursday night tests (and 7:00 PM for the Friday test). Leaving with scratch paper before those times is an academic integrity violation.

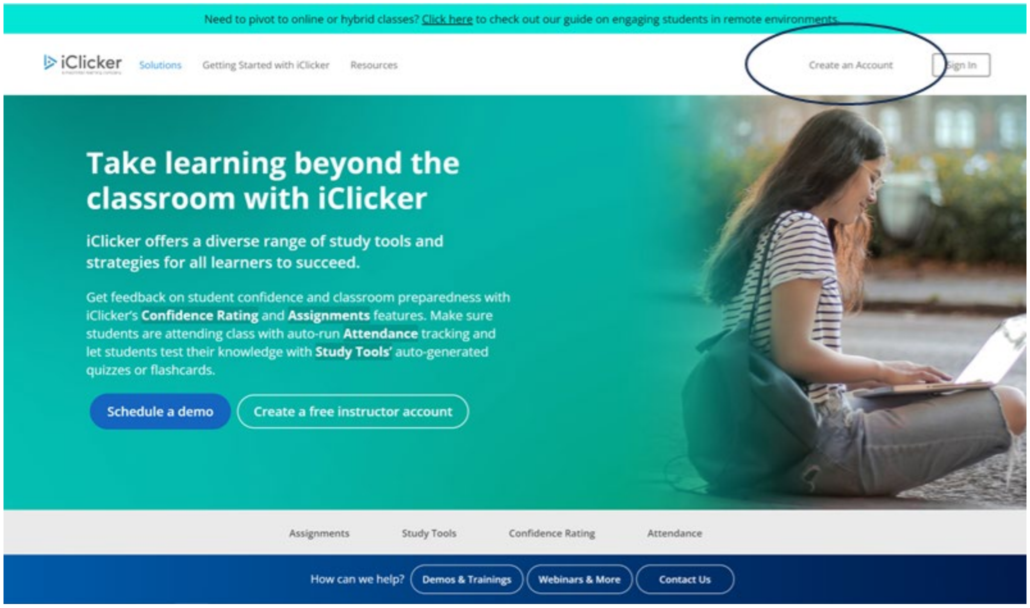
All scratch paper, used or unused, must be turned into the test proctor at the end of the final exam.

Instructions for Registering your iClicker

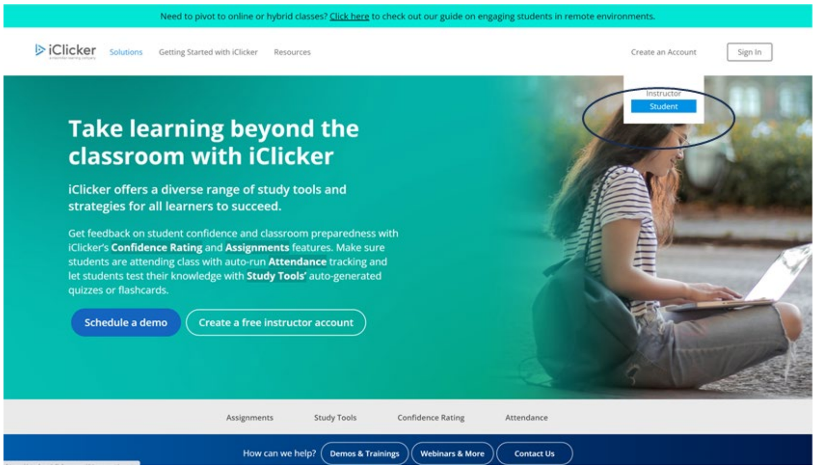
You will use your physical iClicker 2 remote to participate in graded in-class exercises. First, you must register your remote in your iClicker student account. Follow the instructions below to create a student account if you do not already have an iClicker account. Skip to step 8 if you already have an iClicker account.

1. Navigate to iClicker.com

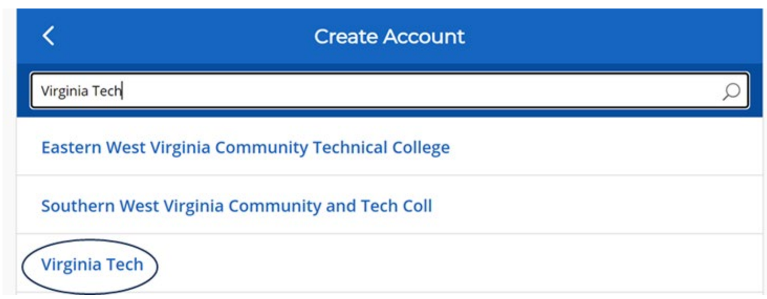
2. Select **Create an Account**



3. Select **Student.**



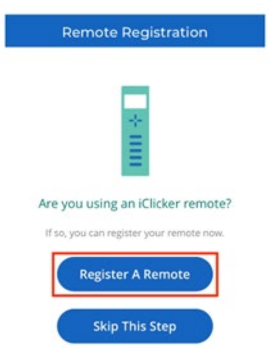
4. Find your Institution (Virginia Tech) and click Next



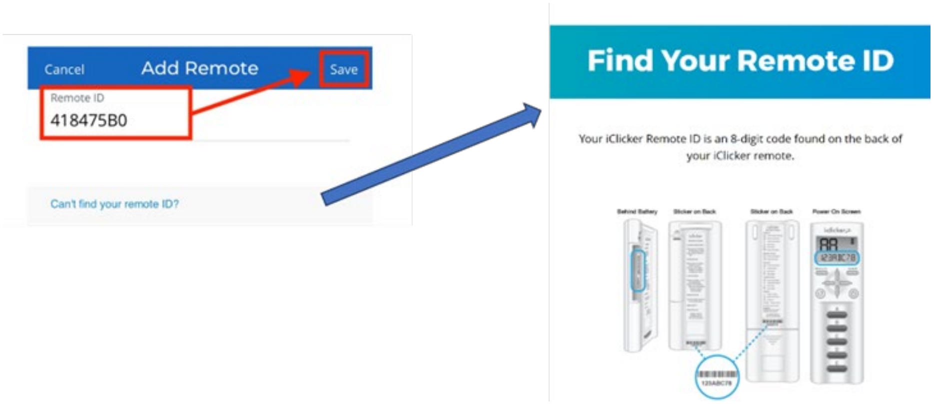
5. Complete the form to create your account and Sign In.

6. Register your remote when you first sign in.

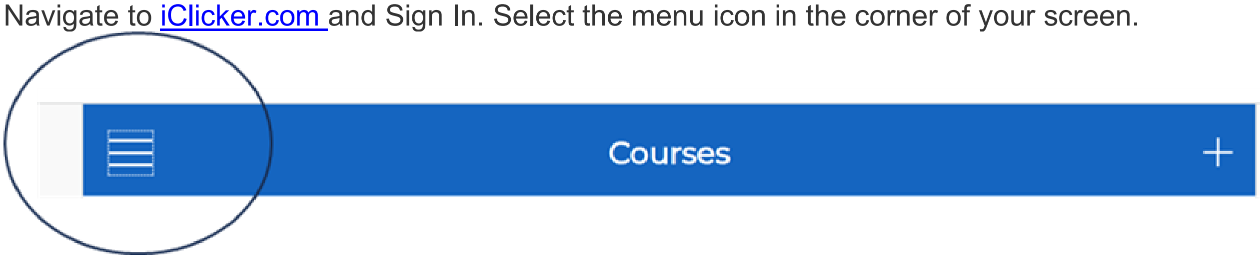
The first time you sign in to the iClicker student app, you will be prompted to add your remote ID. Select **Register A Remote** to continue.



7. Enter your iClicker remote ID and select **Save**. Skip to Step 8.

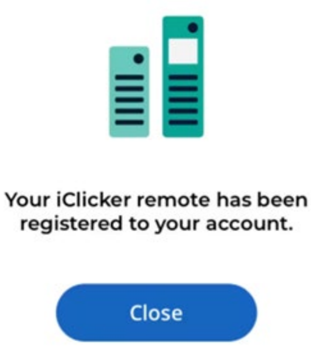


8. If you already have an iClicker student account, you can register your remote in your profile. Navigate to iClicker.com and Sign In. Select the menu icon in the corner of your screen.



Select **Profile, Register Remotes,** Select the plus sign or **Register Remote** (if you have already registered a remote, that information will appear here), and enter your remote ID and select **Save**.

9. After following the instructions above, you will see the following confirmation message: **Your iClicker remote has been registered to your account**.



Next you must make sure you add your instructor's course to your Courses list in the iClicker student app. This will ensure that your iClicker points show up correctly in your instructor's gradebook. <u>Only participation in the course that you are enrolled in will be counted towards your grade.</u>

10. Use the QR code below to complete your iClicker course registration. This code is only good for Dr. Arachchige's Spring 2026 MWF 12:20 PM–1:10 PM CHEM 1036 course.

Read the instructions that came with your clicker so you will be ready to use the clicker in class.

You will probably use an iClicker in multiple classes, either this semester or during other semesters, but you only need one iClicker remote. NOTE: General Chemistry does not use the iClicker Student app, the online version of iClicker, so you cannot respond with a cell phone or computer; you will need a physical iClicker remote. However, as long as you purchase a **new** remote from the campus bookstore or from the Macmillan Student Store and create a student account in iClicker.com, you will gain a 5 year access to the iClicker Student app, at no additional cost. This will allow you to participate in all of your courses without recurring costs, whether it is with a remote, cell phone, or computer, depending on the requirement for that class.

iClicker Grading: The lowest six iClicker performance days (for the MWF class meetings) or the lowest four iClicker performance days (for the MW class meetings) of each student will be dropped. Since this class meets on Mondays, Wednesdays, and Fridays your lowest six iClicker performance days will be dropped. If you miss iClicker questions for any reason (even those excused by the Dean of Students), the zeros received for your absences will be included in the dropped iClicker grades regardless of the reasons. You will receive 1.5 points for giving an answer to an iClicker question and an additional 1 point for providing the correct answer. All general chemistry courses use the base frequency AA.

Instructions for Registering and Accessing the ALEKS online assignments (Homework and Modules). See pages 22-23 for information about the types of ALEKS assignments.

Everyone must have the ALEKS Access Code for our online homework. If you took CHEM 1035 in Fall 2025, you may already have this. Please call 1-800-258-2374 to check if your existing ALEKS Access Code will last you through the Spring Semester. The ALEKS access code gives you access to the homework system and the electronic version of the textbook (e-Book). If you do not have an ALEKS access code, an 18-week access code can be purchased for $81.22 NET directly through Canvas (from the ALEKS website).

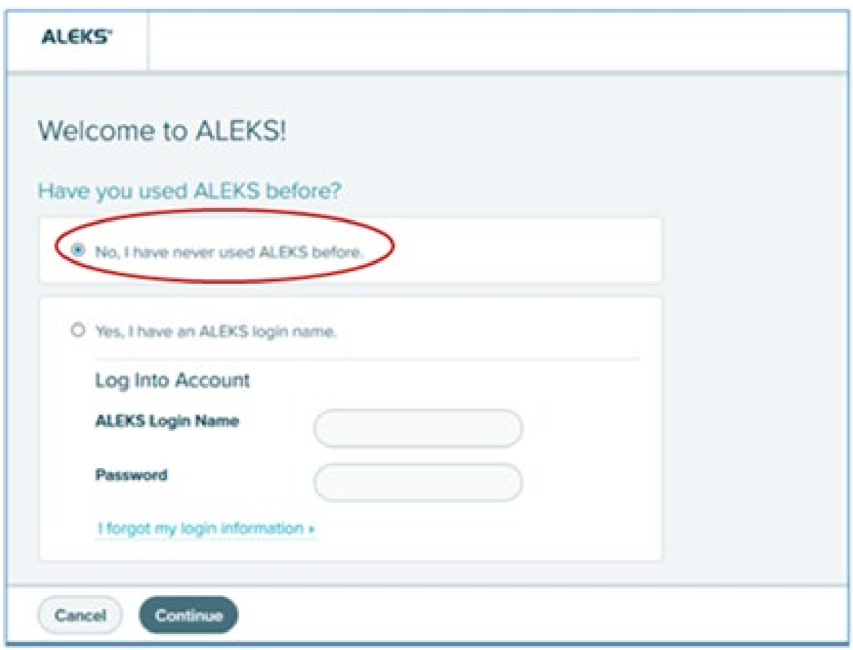
To register in ALEKS:

1. Log into your Canvas account.

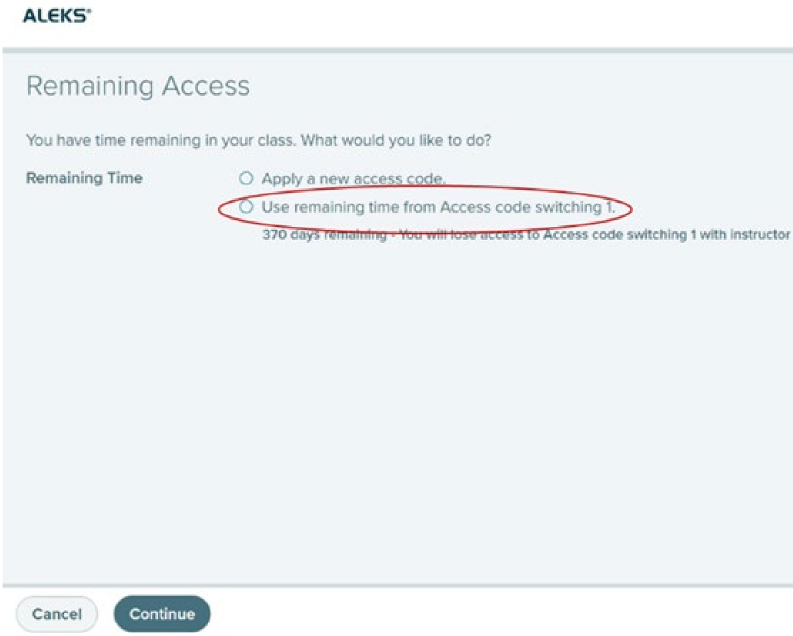
2. Select your course.

3. On the left-side navigation menu, select the tool link ALEKS.

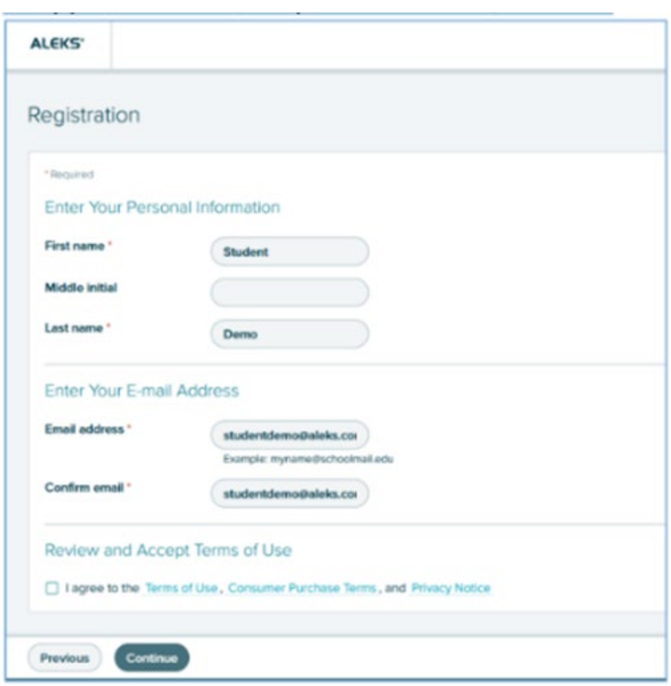
4. If you have never used ALEKS, select the first option, “No, I have never used ALEKS before” and then select Continue.



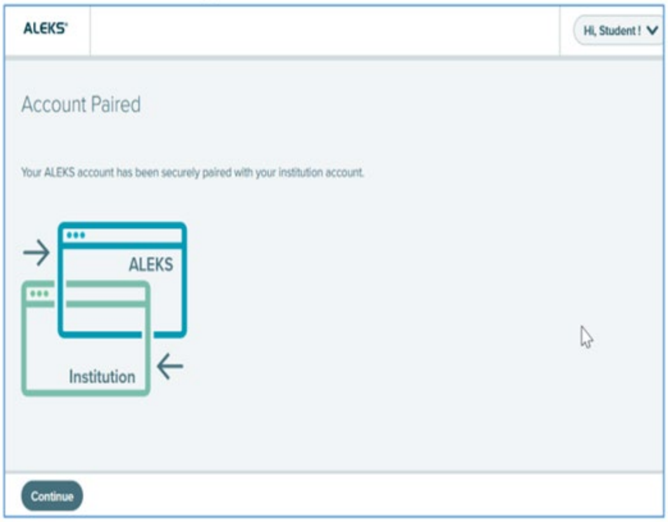
If you already have an ALEKS account, ALEKS will remember your information and you will see the following screen when you select ALEKS in the Canvas homepage. Select “Use remaining time from Access code switching” option. Skip to Step 10.



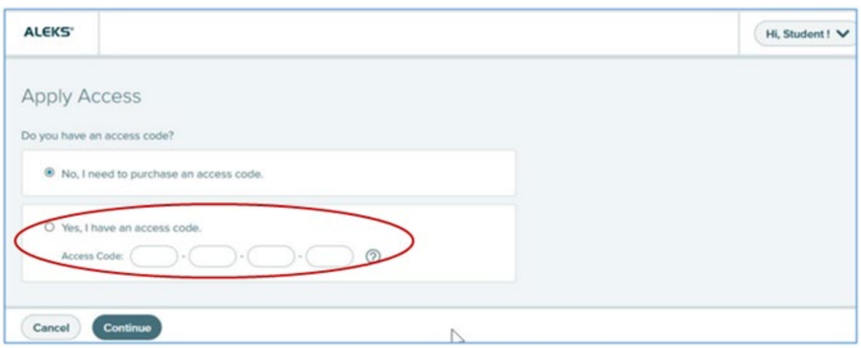
5. Verify your information, accept the Terms of Use, and select Continue. <u>Use only your Virginia Tech e-mail address.</u>



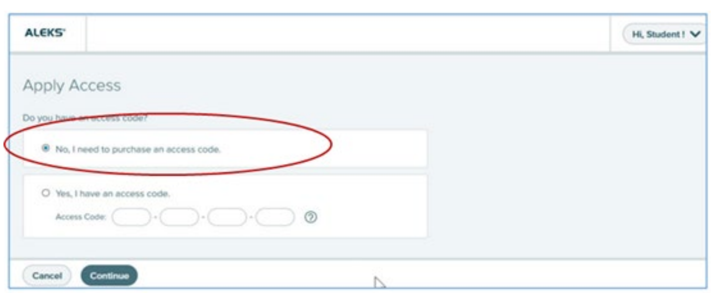
6. Your account is now paired with ALEKS. Select Continue



7. If you purchased a 20-character two-year code from the bookstore, select the second option (circled in red) to continue your registration.



8. If you have not purchased a 20-character access code from the bookstore, select the first option to purchase the code via ecommerce and complete registration. Note that this first option (No, I need to purchase an access code) is the cheaper option.



A two-week financial aid access code for students experiencing financial aid delays:

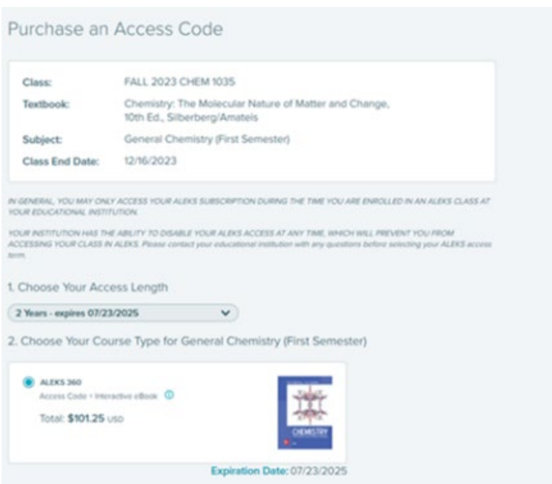
Your class specific Financial Aid Access Code (FAAC) is: F50CB-71183-BDE74-8F610

Use only this 20-character FAAC if you have financial aid delays. The FAAC will provide you with temporary two-week access to the ALEKS course.

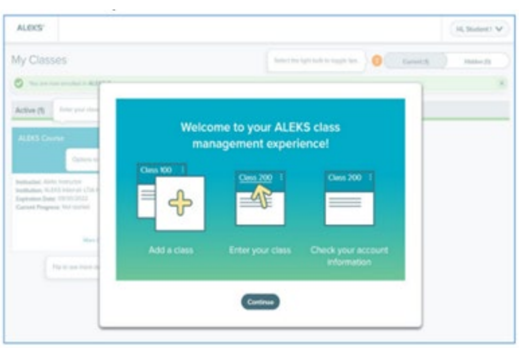
Once the code expires, you will be locked out of your ALEKS account until you purchase a regular Student Access Code. If you enroll in the course using the FAAC, you must purchase a regular Student Access Code before the two-week period ends to extend your account. The start date of your account is the date you activated the FAAC and NOT the date when the account was extended with a purchased Student Access Code. In other words, the FAAC does not add an additional two weeks to your account.

You can extend your access to your new class at any time by selecting "Extend access" from the class tile menu and enter your new access code. Do not create a new ALEKS account to continue your class.

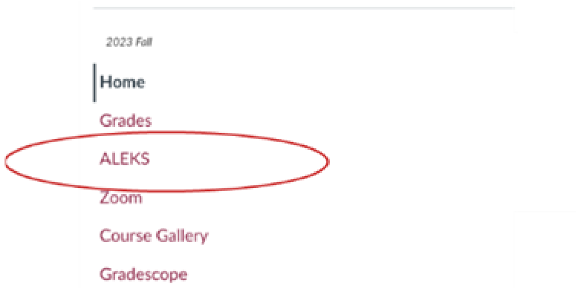
9. Confirm your access code expiration date (two years or 18 weeks). Select Confirm.



10. You are now in your ALEKS Student Account Home.



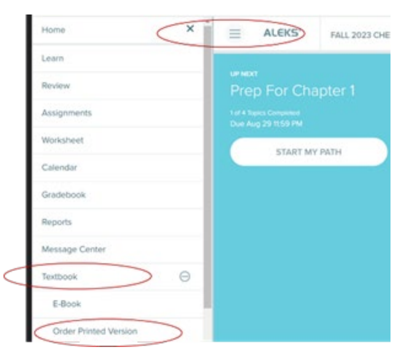
11. When you want to access a homework assignment, **always** choose the ALEKS link from our Canvas home page.



Note: It is your responsibility to keep track of the online assignment deadlines (both Modules and Homework Assignments). Complete all assignments on time. You should check the ALEKS site at least once a day, every day, including on Saturdays and Sundays. Note: ALEKS has its own gradebook. However, only the homework grades transferred to the Canvas gradebook determine your online assignment grade in the course. Therefore, it is also your responsibility to ensure that all online homework grades (for the homework assignments and the modules) have been transferred correctly to the Canvas gradebook upon submission of the assignment. Always check the Canvas gradebook after you submit an assignment. If you encounter any discrepancies and/or technical difficulties, chat or call technical support as soon as possible at https://mhedu.force.com/aleks/s/alekscontactsupport and 1-800-258-2374, respectively.

How to Purchase a Loose-Leaf Copy of the Textbook (Not Mandatory)

Once you purchase ALEKS, you will have access to the electronic version of the textbook, which is all you have to have. But, if you want a hard copy of the textbook, you can purchase a loose-leaf copy of the textbook for $42.00. It comes with three holes punched so you can put it in a ring binder. Purchasing a hard copy of the textbook or not is up to you. Get what will help you be successful. To purchase the loose-leaf version, sign in to our ALEKS site through Canvas. After you complete the Initial Knowledge Check, hit the three dashes that are next to ALEKS on the upper left, open the menu, scroll down to "textbook", open it up and select the option to "order printed version". The book will be mailed to you at the address you specify and it may take 7-10 business days.



Online Homework Assignments

Your lowest five homework grades of the semester will be dropped. If you miss an online assignment (Module or Homework), it will be counted as a zero. The zeros received on the first five assignments will be included as the lowest homework grades. There is no make-up for the first five missed homework assignments, for any reason (including university verified absences). Homework deadlines are not extended for non-university verified reasons.

Two types of assignments will appear on the ALEKS site:

1. Adaptive Modules (Labelled as Primer Chapter 11, Primer for Chapter 12 etc.)
2. Non-Adaptive Homework Assignments (Labelled as Assignment 1 Chapter 11, Assignment 2 Chapters 12 and 13 etc.)

1. *Adaptive Modules (Labelled as Primer Chapter 11, Primer for Chapter 12 etc.)*: These modules are adaptive to your learning. They are mandatory and will be graded. These modules facilitate *independent learning and will typically be due before the lecture.*

<u>Within the first few days of your course</u>, you will take the mandatory **Initial Knowledge Check**. The purpose of this check is to get to know you and to give you "credit" for any of the content of the course you may already know. This "credit" will be different for each student because each student has a unique math background – math courses taken, time since last math class, grades earned in those classes, etc. There is no right or wrong "grade" on the Initial Knowledge Check. Some students come in knowing about 10% of the course content. Comprehending more or less than 10% of the content is perfectly acceptable - you are in this course to learn the material, not to show that you already know it. The result of your Initial Knowledge Check is what ALEKS uses to create your own individualized learning path to work through the mandatory adaptive modules of this course. These modules are labelled as *Primer* for Chapter 11, *Primer for Chapter 12 etc.* You will only have to work through the topics in the adaptive modules that you are most ready to learn. If ALEKS identifies that you know all the topics for the *Primer* for Chapter 11 module when you completed the Initial Knowledge Check, you will earn credit for this module and your Canvas gradebook will be updated with 100% for the module and you can progress to the Primer Chapter 12 module. Similarly, once you complete the Initial Knowledge Check, ALEKS identifies all the topics that you already know in the upcoming adaptive modules. For example, once you have completed the Initial Knowledge Check, your Canvas gradebook may show that you have also earned 100% for Primer Chapter 12. This means that you already also know all the topics tested in this module and you have earned full credit. No further action is necessary for the modules that show up as 100% in your Canvas gradebook. However, upon completion of the initial knowledge check, if your Canvas gradebook shows no score, blank, or any score other than 100% on a particular adaptive module, you still have to complete those modules before the deadline and learn the remaining topics to earn the remaining credit. After the due date you are still able to complete the incomplete topics, but your grade will not change. During the adaptive modules, you can access the electronic textbook anytime to read information that will help you answer each question.

2. *Non-Adaptive Homework Assignments (Labelled as Assignment 1 Chapter 11, Assignment 2 Chapters 12 and 13 etc.)*: These assignments are mandatory and will be graded. In addition, to the adaptive modules, you will also complete assignments that are not part of your adaptive learning experience. These non-adaptive assignments will not affect your adaptive learning path and must be completed separately in addition to the adaptive modules. THE **HIGHEST** SCORE

RECEIVED FOR EACH ASSIGNMENT WILL BE TAKEN, NOT THE AVERAGE SCORE. THERE IS NO PENALTY FOR ATTEMPTING AN ASSIGNMENT MULTIPLE TIMES. If you miss 3 out of 10 questions on your first submission, you only have to work the 3 missed problems on your second submission – you do NOT have to work all 10 problems again! After the due date for each non-adaptive assignment, you can access the assignment as a *Review Assignment* for practice. Your original grade on the assignment does not change. You can also access the electronic textbook to read information that will help you answer each question during the non-adaptive assignments.

Detailed Course Schedule for CHEM 1036

Dates	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Week 1 Jan. 18-24		No Class		Syllabus/ What you should already know REQUIRED Worksheet Week 1 (All Worksheets are on Canvas)	1. Register for ALEKS 2. Register your iClicker 3.Read pp. 444-451 REQUIRED STUDIO MEETS	**Chapter 11** Hybridization (sp, sp^2, sp^3)	REQUIRED Start Worksheet 2
Week 2 Jan. 25-31	Read pp. 452-454	**Chapter 11** Sigma and Pi Bonding	Read pp. 471-475 REQUIRED Primer Chap 11	**Chapter 12** Introduction to attractive forces. Types and strengths of intermolecular forces. Required Assignment 1 Chapter 11	REQUIRED STUDIO MEETS/QUIZ	**Chapter 12** Introduction to attractive forces. Types and strengths of intermolecular forces. Read pp. 486-492	REQUIRED Start Worksheet 3
Week 3: Feb 1-7	Read pp. 535-540	**Chapter 12** Types and strengths of intermolecular forces. REQUIRED Primer Chap 12 and Chap 13 Part I	Read pp. 546-551	**Chapter 13** Intermolecular forces and solubility Read pp. 552-555 REQUIRED Primer Chap 13 Part II	REQUIRED STUDIO MEETS/QUIZ	**Chapter 13** Factors that determine solubility	REQUIRED Start Worksheet 4

Week 4: Feb 8-14	REQUIRED Primer Chap 13 Part III	**Chapter 13** Heat of solution, heat of hydration, temperature and pressure effects on solubility	Read pp. 555-560 REQUIRED Primer Chap 13 Part IV	**Chapter 13** Concentration units Required Assignment 2 Chapters 12 and 13	REQUIRED STUDIO MEETS/QUIZ	**Chapter 13** Colligative properties	REQUIRED Start Worksheet 5
Week 5: Feb 15-21	Read pp. 560-570 REQUIRED Primer Chap 13 Part V	**Chapter 13** Colligative properties REQUIRED Assignment 3 Chap. 13		**Chapter 16** Introduction to reaction rate Rate laws; rate constant, reaction order REQUIRED Assignment 4 Chap. 13	REQUIRED STUDIO MEETS/QUIZ REQUIRED **TEST 1** (Chapters 11,12, and 13)	**Chapter 16** Integrated rate laws Read pp. 702-719	REQUIRED Start Worksheet 6
Week 6: Feb 22-28	REQUIRED Primer Chap 16 Part I Read pp.720-726	**Chapter 16** Collision theory, activation energy, Arrhenius equation	REQUIRED Primer Chap 16 Part II	**Chapter 16** Collision theory, activation energy, Arrhenius equation REQUIRED Assignment 5 Ch. 16	REQUIRED STUDIO MEETS/QUIZ	**Chapter 17** Equilibrium reactions; equilibrium constant	REQUIRED Start Worksheet 7

Week :7 March 1-7	Read pp. 753-776	**Chapter 17** K_c vs K_p; reciprocal and coefficient rules REQUIRED: Assignment 6 Chap. 16	Read pp. 777-788 REQUIRED Primer Chap 17 Part I	**Chapter 17** Equilibrium Calculations REQUIRED Primer Chap 17 Part II	REQUIRED STUDIO MEETS/QUIZ	**Chapter 17** LeChatelier's Principle	REQUIRED Start Worksheet 8
	SPRING BREAK	SPRING BREAK	SPRING BREAK	SPRING BREAK	SPRING BREAK	SPRING BREAK	SPRING BREAK
Week 8: Mar 15-21	Read and pp.803-809	**Chapter 18** Arrhenius and Bronsted-Lowry definitions of acids and bases REQUIRED Assignment 7 Chap. 17	Read pp. 809-813 REQUIRED Pathways Assignment - Equilibrium	**Chapter 18** pH and pOH scales REQUIRED Assignment 8 Chap. 17	REQUIRED STUDIO MEETS/QUIZ REQUIRED **TEST 2** (Chapters 16 and 17)	**Chapter 18** Strong acids and bases; weak acids; K_a constant	REQUIRED Start Worksheet 9

Week 9: Mar 22-28	Read pp. 813-817	**Chapter 18** Factors affecting acid strength; weak bases Read pp. 818-824 REQUIRED Primer Chap 18 Part I	Read pp. 825-832 REQUIRED Primer Chap 18 Part II	**Chapter 18** Weak acid and weak base calculations; polyprotic acids REQUIRED Assignment 9 Chap. 18	REQUIRED STUDIO MEETS/QUIZ Read pp. 833-836	**Chapter 18** Acid-base properties of salt solutions REQUIRED Pathways Assignment-Acidic Seas	REQUIRED Start Worksheet 10
Week 10: Mar 29-Apr 4	Read pp. 853-863	**Chapter 19** Buffer solutions REQUIRED Primer Chap 18 Part III	Read pp. 872-882	**Chapter 19** Buffer calculations REQUIRED Assignment 10 Chap. 18	REQUIRED STUDIO MEETS/QUIZ Read pp. 863-866	**Chapter 19** Strong acid-strong base titrations Read pp.866-871	REQUIRED Start Worksheet 11

Week 11: April 5-11		**Chapter 19** Weak acid/strong base and weak base/strong acid calculations REQUIRED Primer Chap 19 Part I		**Chapter 19** Titration calculations REQUIRED Assignment 11 Chap. 19	REQUIRED STUDIO MEETS/QUIZ	**Chapter 19** Titration calculations REQUIRED Primer Chap 19 Part II	REQUIRED Start Worksheet 12
Week 12 April 12-18	Read pp. 907-918	**Chapter 19** Acid-base indicators; solubility equilibria REQUIRED Assignment 12 Chap. 19		**Chapter 20** Reaction spontaneity; entropy REQUIRED Assignment 13 Chap. 19	REQUIRED STUDIO MEETS/QUIZ REQUIRED **TEST 3** (Chapters 18 and 19)	**Chapter 20** Enthalpy, entropy, and spontaneity; Gibbs' free energy REQUIRED Primer Chap 20 Part I Read pp 918-928	REQUIRED Start Worksheet 13
Week 13 April 19-25		**Chapter 20** Enthalpy, entropy, and spontaneity; Gibbs' free energy REQUIRED Assignment 14 Chapter 20	REQUIRED Primer Chap 20 Part II Read pp. 928-940	**Chapter 20** Temperature, free energy, and equilibrium constant REQUIRED Assignment 15 Chap. 20	REQUIRED STUDIO MEETS/QUIZ REQUIRED Primer Chap 21 Part I	**Chapter 21** Introduction to Redox reactions and Balancing oxidation-reduction reactions REQUIRED Primer Chap 21 Part II	REQUIRED Start Worksheet 14

Week 14 April 26-May 2	REQUIRED Pathways Assignment Flint Water Crisis	**Chapter 21** Voltaic cells: construction, standard notation, Read pp. 962-970 REQUIRED Assignment 16 Ch. 21	REQUIRED Pathways Assignment Molecular Art	**Chapter 21** Cell potential; Table of half-cell reduction potentials	REQUIRED STUDIO MEETS/QUIZ REQUIRED Assignment 17 Chapter 21	**Chapter 21** Cell potential calculations REQUIRED **TEST 4 (Chapters 20 and 21)**	REQUIRED Start Worksheet 15
Week 15 May 3-9	Read pp. 971-976 REQUIRED Primer Chap 21 Part III	**Chapter 21** Gibbs Free energy and equilibrium constant; Nernst equation		**Chapter 21** Gibbs Free energy and equilibrium constant; Nernst equation REQUIRED Assignment 18 Chapter 21			REQUIRED **Cumulative Final Exam:** Chapters: 11, 12, 13, 16,17,18, 19, 20, 21)